getting a larger training set, we're getting larger databases and we're retraining our network. And that's going to be a "forever" kind of operation. We're going to continue to build up our training infrastructure, data centres and how we're gathering all the data. There's like a whole data architecture of becoming a production-grade AI training company that we have to do in order to do a good job with XeSS. That's the infrastructure that we're building now and that's a year-long effort. So I think of it as, the XeSS algorithm being the first of what I expect to be an entire family of AI-based render techniques. We don't know what those are going to be yet but we do know that AI is basically disrupting everything and we'd be poor stewards of graphics if we didn't look for avenues to apply AI.

When you mentioned that you're looking at a really large dataset to train the neural net, is there some way of optimising what would be a better subject or data set to use? Let's take the graphics already present in current gen games as an example. Training on that would definitely give you short-term benefits but if you picked a wider data-set then that might help you with other applications, perhaps in some other Intel IP. So what's the focus area going to be like?

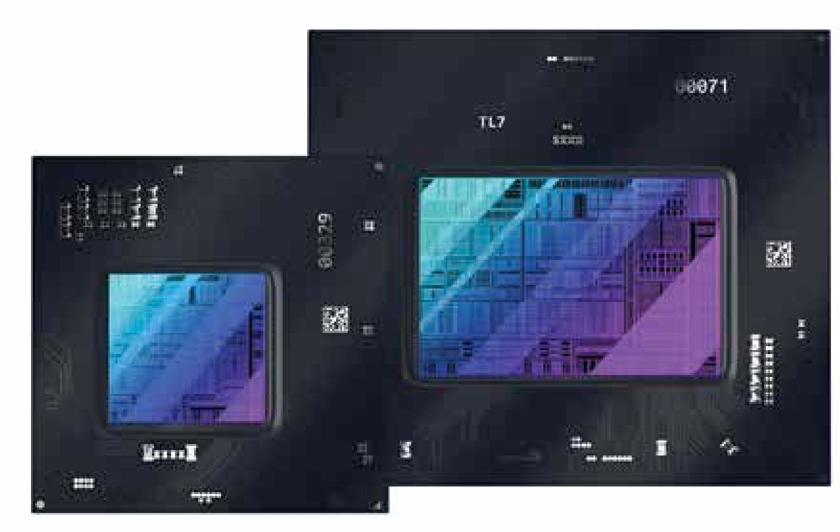
The focus is right now on the titles that we do target. We're building our dataset with multiple titles but this is the whole art of data science. I'm not sure how much time you've spent with the Googles and the Amazons and the people who've been doing all the data science work related to training algorithms. It's about how you craft that dataset, how you maintain that data set and how you efficiently pull that data into the training processes. It's like rocket science. It's like art and is evolving over time. It's a large investment trying to figure out if we're using synthetics or if we're using benchmarks or if we're using actual gameplay. And how often do we reset our training set? It's all a massive process that we're really beginning to appreciate the magnitude of.

Is there some sort of cadence that you're looking to come out with newer XeSS profiles? Would you be updating with each zero-day game drivers or is it a six-month cycle?

Much lower than that. It's more like 6 to 12 months but it's not quite that slow. Our XeSS SDK is updating on a cadence that's relatively moderate. You don't want your SDK to change very often. And we're going to be updating network weights with almost every game integration. Think of it as, when we're doing work with a game engine, that's when we have an opportunity to provide a new network. Because most of these game guys don't want a random network to show up later. They want the network to remain stable so that they can do their verification. They'd want to know that their game is not going to perform poorly because there were issues with the training. So they'd want a network that they can ship their game with. That means that our opportunity to update the network is roughly around the time they're performing integration into the games.

On mobile GPUs where there is a certain dynamic power envelope within which the GPU has to function, in such cases, what does the algrithm optimise? Will it drop the FPS or will it be an image quality reduction?

That's an area where we haven't done much yet which is, how do we want to dynamically adjust the render behaviour based on the power profile. Today, it's all statically configured. And you'll have a network topology that can be configured to high, medium or low within the XeSS settings. And then you may get a sub-par experience if you run into power issues and you might have been better served by having a higher resolution render and not using XeSS. It's very much a discovery. That's something we'd like to figure out going forward - how we can help users either understand what choices they should make (by giving references) or by doing it dynamically. That's the best case. To see what's happening with the GPU load, temperatures and TDPs, so that we can shift energies and change the settings. There's actually work done in the industry to see if we can get a common API within the game engine that would allow us to give hints to the game engine to lower the power profile as per external signals. Right now that doesn't exist so there's lots of work to be done to improve the experience for end-users on how power, quality and frame rates are all balanced.



Intel Arc GPUs